

Artificial Intelligence Lab – Answers

Experiment 1

1. Artificial Intelligence (AI) is a branch of computer science that deals with designing intelligent machines that can perform tasks that normally require human intelligence such as learning, reasoning and problem solving.
2. AI components include machine learning, natural language processing, knowledge representation, reasoning and computer vision.
3. Categories of AI: Reactive Machines Limited Memory Theory of Mind Self Aware AI
4. Foundations of AI include mathematics, psychology, neuroscience, philosophy and computer engineering.
5. Applications of AI include healthcare, robotics, autonomous vehicles, fraud detection and recommendation systems.

Experiment 2

1. Types of task environments: Fully observable vs partially observable Deterministic vs stochastic Static vs dynamic Discrete vs continuous Single agent vs multi agent Known vs unknown
2. PEAS for tennis against wall Performance: Accuracy of hitting the ball and number of successful returns. Environment: Tennis court, wall and ball. Actuators: Hands, legs and racket. Sensors: Eyes to observe the ball.
3. 8 Puzzle state space formulation The 8 puzzle consists of eight numbered tiles and one blank space. The objective is to reach the goal configuration by sliding tiles into the blank space. Each configuration represents a state and each move represents an operator.

Experiment 3

1. Problem solving agent is an intelligent agent that searches for a sequence of actions to reach a goal state.
2. Uninformed search algorithms include BFS, DFS, Uniform Cost Search, Depth Limited Search and Iterative Deepening Search.
3. Iterative deepening search repeatedly performs depth limited search with increasing depth until the goal is found.

Experiment 4

1. Uninformed search does not use domain knowledge while informed search uses heuristics to guide the search.
2. Greedy best first search uses heuristic function $h(n)$ which estimates the distance from current node to goal node.
3. In 8 puzzle admissible heuristics include misplaced tiles and Manhattan distance.
4. A* algorithm uses evaluation function $f(n)=g(n)+h(n)$ and it is complete and optimal.
5. Memory bounded heuristic search algorithms include IDA and SMA.

Experiment 5

1. Adversarial search is used in competitive environments such as games where one player's gain is another player's loss.
2. A game tree represents all possible moves in a game.
3. Pruning reduces the number of nodes evaluated in a game tree.
4. Complexity of alpha beta pruning in best case is $O(b^{d/2})$.
5. Minimax value of the tree is 3.

Experiment 6

1. Rules in Prolog define relationships between facts. Example: `mother(X,Y):-parent(X,Y),female(X).`
2. Facts are statements that describe true relationships. Example: `parent(john,mike).`
3. Command to load file: `consult(filename).`
4. Goals in Prolog are queries asked to the knowledge base. Example: `?- parent(john,X).`
5. The query `p(X,Y)` will produce infinite answers because of recursive rule `p(X,Y):-p(Y,X).`

Experiment 7

1. Planning in AI is the process of finding a sequence of actions to transform initial state into goal state.
2. Types of planning: Classical planning Non classical planning

3. Partial order planning allows actions to be executed in flexible order rather than fixed sequence.
4. PDDL is a language used to represent planning problems.
5. Hierarchical planning divides a complex problem into smaller subproblems.

Experiment 8

1. Bayesian belief network is a probabilistic graphical model that represents variables and their conditional dependencies using directed acyclic graph.
2. Applications include medical diagnosis, speech recognition and bioinformatics.
3. Advantages include handling uncertainty, probabilistic reasoning and modeling causal relationships.
4. Bayesian networks represent probabilistic relationships among variables.
5. Example: Rain, Sprinkler and Wet Grass where wet grass depends on rain and sprinkler.